

Incredible S

HTC launches Incredible S an Android phone in India, which will soon receive a Gingerbread update

Chrome gets new face

Google chrome is getting a new icon which embodies the Chrome Spirit

Inside your phone

ARM and System-on-a-Chip (SoC) is where the action currently is. We take a look at the players and what they're offering, and the game itself...

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This year promises to be a rich one for mobile processing. For while the 1 GHz barrier was crossed back in 2008, the future promises more cores, higher clock speeds, and best of all – at no loss to battery life. ARM is by far the most preferred platform till date.

The big players that are making waves in the SoC (System-on-a-Chip) arena are Qualcomm, Samsung, NVIDIA, Texas Instruments, Apple, ST-Ericsson. There are a few others as well such as Broadcom, Fujitsu, Infineon Technologies, Sharp, Cirrus Logic, to name a few, but we'll focus on the active ones.

Qualcomm: Snapdragon

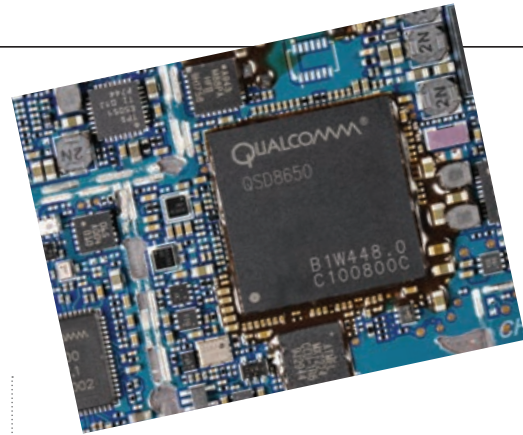
One of the most widespread SoC (*plural: Systems-on-a-Chip*) is the "Snapdragon", which is Qualcomm's codename for a family of SoC designs. These SoC are divided into generations in order to differentiate. Their Scorpion CPU is home grown, but based roughly

on the ARM Cortex-A8. The Adreno GPU inside this SoC is made by Qualcomm, but courtesy Imageon, which was AMD's mobile graphics division, that Qualcomm bought off.

The second generation of Snapdragons are made on a 45nm process and have an improved graphics subsystem that is supposed to be three times more powerful than its predecessor, and this SoC is claimed to consume 33 per cent less power.

The third generation of Snapdragon SoCs feature the Adreno 220, a graphics part that is three times more powerful than the Adreno 205. Additionally, this SoC can be a dual core part, meaning two Scorpion cores. Evidently, dual core designs consume less power if applications are coded to take advantage of multiple threads.

The fourth generation of Qualcomm's Snapdragon SoC has just been announced and is built around a 28nm fab. Qualcomm is going with a new CPU called Krait, although details of the architecture are scarce. Out of order (OoO) execution is supported, and these SoC will be primarily dual core, although single and quad core



variants are available too. The 3xx series of Adreno GPUs also warrant mention, for Qualcomm is claiming similar processing power as the XBOX 360 and PS3.

Qualcomm has ditched the QSD moniker for marketing reasons, and now cellphone SoC will be called MSM, while for tablets, the APQ designation will be used.

At the moment, Qualcomm is about the most preferred SoC vendor in the market, additionally, they are the only certified SoC manufacturer for Windows Phone 7, although this might change pretty soon.

NVIDIA: Tegra, Tegra 2

NVIDIA's SoC started off with their Tegra chip, which wasn't too hot, but it served as the precursor to Tegra 2, which is definitely creating waves.

Tegra was based around a single core ARM 11 design and wasn't scalable. As a result, it didn't do too well. At around the same time NVIDIA was taking a beating

Year	2008		2009			2010		
SoC name	Snapdragon (1st Gen)	Tegra	Snapdragon (2nd Gen)	Hummingbird	OMAP3	Tegra 2	OMAP4	
Company Name	Qualcomm	NVIDIA	Qualcomm	Samsung	Texas Instruments	NVIDIA	Texas Instruments	
Instruction set	ARMv7a	ARMv7a	ARMv7a	ARMv7a	ARMv7a	ARMv7a	ARMv7a	
CPU codename	Scorpion	ARM 11	Scorpion	Hummingbird	ARM Cortex-A8	Tegra 2	Cortex-A9	
CPU architecture	ARM Cortex-A8	ARM 11	ARM Cortex-A8	ARM Cortex-A8	ARM Cortex-A8	ARM Cortex-A9	Cortex-A9	
No. of Cores	1	2	1	1	1	1, 2	2	
CPU clock speeds	up to 1 GHz	600 MHz - 800 MHz	800 MHz - 1.3 GHz	1 GHz	600 MHz - 1.2 GHz	Up to 1.2 GHz	1 GHz - 1.5 GHz	
GPU	Adreno 200	GeForce ULP	Adreno 205	PowerVR SGX535/540	PowerVR SGX530	8 core GeForce ULP	PowerVR SGX540	
Common SoC model numbers	QSD8250, QSD8650	Tegra APX 2500 / 2600, Tegra 650/600	QSD 8250A / 8650A, MSM 7230 / 8255 / 8655	S5PC110, S5PC100	OMAP 3410 / 3430 / 3440 / 3630 / 3640	Tegra 250, Tegra 250 3D,	OMAP4430, OMAP4440	
Cellphones based on this SoC	Dell Venue Pro, Acer Liquid, HTC HD7, HTC Desire, Google Nexus One, Sony Ericsson X10i, HTC Evo 4G	Microsoft Zune	HTC Desire Z, HTC Evo Shift 4G, HTC Desire HD, Sony Ericsson XPERIA Duo, Sony Ericsson XPERIA Arc	Samsung Galaxy S, Samsung Wave, Google Nexus S, iPhone 4 (modified slightly)	Motorola Flipout, Charm, Milestone, XT720, Delfy, Droid X, Nokia N900, Palm Pre, Samsung i8910, i9003	LG Optimus 2x, Motorola Atrix 4G	-	
Tablets based on this SoC	Dell Streak	Mobinova Elan	-	Apple iPad (modified A4 core)	Beagle Board, Touch Book, Pandora	Samsung Galaxy Tab 10.1, ASUS Slider, Notion Ink Adam Tablet	RIM Blackberry Playbook, PandaBoard, LG Optimus 3D	
Interesting facts	The first SoC that powered capacitive touch devices and made a name for itself, and it is one that continues to power devices even today	Tegra wasn't the hit NVIDIA wanted it to be, but it got them started on the SoC road, from where they haven't looked back	Better performance at no loss in battery life is the selling point here	Samsung's first SoC was a hit, unlike NVIDIA and some of the best selling Android phones, including the Galaxy S are based on it	The OMAP3 has the most number of SoC models within the family, with some 9 variants. It has seen extensive use with Motorola-based handsets	One of the more exciting SoC around, the Tegra 2 is all set to make a splash later in 2011 as more and more devices come in	Another powerful SoC that hasn't seen a lot of takers, but we're hoping that changes for with the A9 design, TI potentially has a winner on their hands	